

Requirements Document (PRD)

Hospital Management System

1. Introduction

The hospital wishes to implement a software system that allows centralizing the management of the institution.

The main objective is to improve internal organization, reduce administrative errors, and facilitate access to the information necessary for the patient's care.

The system must integrate clinical and administrative information, allowing the management of different hospital processes.

2. Scope of the system

The system must cover the following processes:

- Patient management
- Medical appointment administration
- Electronic medical record
- Medical staff management
- Billing and service collection
- Medication and supplies inventory control
- Report and statistics generation

3. System Actors

The system will be used by different types of users, each one with specific responsibilities within hospital management:

- Administrative staff: User responsible for patient management, medical appointment administration, and billing and collection processes.
- Doctor: User in charge of medical care for patients and of recording and consulting the electronic medical record.
- Pharmacist: User responsible for managing the inventory of medications and supplies, as well as recording medication delivery.
- Analyst: User in charge of consulting and analyzing system reports and statistics.
- Patient: User who can consult their personal information and their electronic medical record, as well as receive notifications related to appointments and billing.

4. Functional requirements

4.1 Patient management

The system must allow:

- Register patients with personal data and medical background.
- Assignment of a unique identifier to each patient.
- Modify the personal and contact details of a registered patient.
- Record medical background (such as blood type, allergies, chronic diseases, and regular medication).
- Search for patients by name, identity document, or identifier number.
- Visualize a patient's complete information.

4.2 Appointments and medical schedule

The system must allow:

- Creating medical appointments by associating the patient, specialty, and professional.
- Visualize the availability of registered doctors.
- Avoiding the assignment of overlapped appointments.
- Modify the date and time of an existing appointment.
- Cancel previously assigned appointments.
- Notify the patient when an appointment is modified or canceled.
- Manage the status of appointments throughout their lifecycle.

Appointments may be found, among others, in the following states:

- Confirmed
- Canceled
- Attended
- Not attended

4.3 Electronic clinic history

The system must have an electronic clinic history system that allows:

- Register diagnoses, treatments, and patient's clinical progress.
- Users must be able to edit their electronic clinical record.
- Each record done in the electronic clinical record must have an associated date.
- Restrict the access to clinical information only to authorized users.
- Allow that patients consult their electronic clinical records, but it can't be modified by them.

4.4 Medical staff management

The system must allow:

- Register medical professionals with their personal and contact details.
- Association of medical specialties with each professional.
- Definition and modification of attention hours.
- Activate or deactivate professionals for your availability in the agenda.
- Manage roles and access permissions to system functionalities.
- Validate permissions before executing actions.

4.5 Billing and collection

The system must allow:

- Generating electronic invoices for medical consultations.
- Generating electronic invoices for the delivery of medications.
- Associating invoices to patients and the corresponding services.
- Assigning a unique identifier to each invoice.
- Sending automatic notifications to patients when an invoice is generated.
- Attach the invoice in digital format to the notification sent.

4.6 Inventory of medicines and supplies

The system must allow:

- Register medications and supplies with identification information, available stock, and expiration date.
- Automatically update stock when registering medication deliveries.
- Detect minimum stock levels.
- Notify medications or supplies about to expire.
- Consult inventory sorted by available quantity or expiration date.

4.7 Reports and statistics

The system must allow generating reports that include, at least:

- Number of consultations by specialty.
- Appointment status.
- Financial reports income.
- Medication and supplies consumption.
- Filters by date ranges for all reports.

5. General considerations

- Date validations must prevent invalid ranges.
- Notifications must be sent within a reasonable time after the events that trigger them.
- Each patient's identity document and identifier should be unique.